

CMP-07 - Judgment Service

Produce evidence-linked Decision Objects that expose alternatives, uncertainty, values, authority context, rationale, and review conditions.

Authority and state ownership

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 09. Authoritative state: Decision Objects, alternative evaluations, uncertainty assessments, review conditions. Explicitly non-owned: Intent, authority grants, evidence custody, action execution.

Critical invariant

A Decision Object records judgment but does not itself create Permission, Authority, or a real-world effect.

Logical interfaces

CMP-07-IF-01 / Command / Request Judgment

Command `Request Judgment` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-07-IF-02 / Command / Record Decision

Command `Record Decision` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-07-IF-03 / Command / Reconsider Decision

Command `Reconsider Decision` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-07-IF-04 / Command / Withdraw Decision

Command `Withdraw Decision` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-07-IF-05 / Query / Get Decision Object

Query `Get Decision Object` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-07-IF-06 / Query / Explain Rationale

Query `Explain Rationale` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-07-IF-07 / Query / List Alternatives

Query `List Alternatives` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-07-IF-08 / Event / Decision Recorded

Event `Decision Recorded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-07-IF-09 / Event / Decision Reconsidered

Event `Decision Reconsidered` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-07-IF-10 / Event / Decision Withdrawn

Event `Decision Withdrawn` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Proposed > Deliberating > Decided > Under Review > Superseded or Closed. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: missing alternative, hidden uncertainty, authority laundering, stale evidence, irreproducible rationale. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-07-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-07-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-07-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-07-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-07-TST-005 - Critical invariant. A Decision Object records judgment but does not itself create Permission, Authority, or a real-world effect. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-07-TST-006 - Failure containment. Inject missing alternative, hidden uncertainty, authority laundering, stale evidence, irreproducible rationale. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-07-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-07-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-07-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-07-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.